



## Screenshot 2026-05-14 at 8.49.22 PM.jpg

I found several relevant mechanistic-interpretability leads, but I did not find a clean set of results that fully matches every one of your five categories for the exact target of “where the final answer becomes determined inside the forward pass of a reasoning model” and the “self-corrects on regeneration” phenomenon. The closest matches are mostly partial: step-level logit-lens analyses, activation-patching studies on reasoning tokens, and a few mechanistic studies of distilled reasoning models.<sup>[1]</sup> <sup>[2]</sup> <sup>[3]</sup>

### What I could verify

- **When Reasoning Traces Become Performative: Step-Level ...** (2026-05-12) is a strong match for your category 1 and partly 3: it uses **logit lens / tuned lens** on intermediate CoT positions to track answer commitment, and it explicitly measures when the answer becomes decodable during reasoning traces.<sup>[1]</sup>
- **Do Latent-CoT Models Think Step-by-Step? A Mechanistic Study on ...** (2026-01-30) is a good match for category 1 and 2 in a different setting: it uses **logit lens, linear probes, and targeted activation patching** on a sequential reasoning/distillation model, but it is about **latent-CoT / CODI**, not the exact self-correcting regeneration behavior you described.<sup>[2]</sup>
- **Empirical, Attention-Based and Mechanistic Insights into Distilled DeepSeek R1 Models** (2025-09-27) is a good match for category 2 and 4: it uses **activation patching** on DeepSeek-R1 distilled models and shows that patching specific reasoning-token activations can flip answers, but it does not look like a direct study of the regeneration/self-repair mechanism you asked about.<sup>[3]</sup>

### Closest partial matches

| Work                                                      | Relevant methods                                                       | Models tested                                                    | Behavioral only or mechanistic                                                   | Main finding                                                                                        | What it leaves open                                                                     |
|-----------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| When Reasoning Traces Become Performative: Step-Level ... | Logit lens, tuned lens, Patchscopes, direction ablation <sup>[1]</sup> | DeepSeek-R1-distilled variants and matched models <sup>[1]</sup> | Mechanistic, because it inspects internal activations/projections <sup>[1]</sup> | Answer commitment tends to emerge late and can be tracked step-by-step in CoT traces <sup>[1]</sup> | It does not by itself identify a specific repair/self-correction circuit <sup>[1]</sup> |

| Work                                                                                  | Relevant methods                                                       | Models tested                             | Behavioral only or mechanistic | Main finding                                                                                                                  | What it leaves open                                                                                      |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Do Latent-CoT Models Think Step-by-Step? A Mechanistic Study on ...                   | Logit lens, linear probes, attention analysis, activation patching [2] | CODI latent-CoT teacher-student model [2] | Mechanistic [2]                | Internal state often contains the correct intermediate state, but the model can still rely on late-stage fusion/shortcuts [2] | It is not about the exact “edited reasoning sentence regenerates back to the same answer” phenomenon [2] |
| Empirical, Attention-Based and Mechanistic Insights into Distilled DeepSeek R1 Models | Activation patching, residual-stream patching, logit difference [3]    | R1-Llama-8B, R1-Qwen-7B, R1-Qwen-1.5B [3] | Mechanistic [3]                | Patching reasoning-token activations can substantially affect the final answer, including answer-flipping effects [3]         | It does not isolate a dedicated self-repair circuit or regeneration-specific correction mechanism [3]    |

## What I did not find

- I did not find a verified 2024–2026 paper/post that directly proves a specific “repair/self-correction circuit” responsible for pulling a perturbed reasoning trace back to the committed answer in the exact sense you described.
- I also did not find a clear result that specifically studies editing intermediate reasoning sentences and then measuring whether regeneration self-corrects back to the same final answer across the exact models you listed, at least not in a way I can safely cite from the retrieved sources.
- I did not find a definitive study that cleanly combines all of: long CoT, intermediate-coordinate decodability, causal commit-point localization, token-forcing interventions, and explicit self-correction under regeneration in one paper/post from the sources retrieved here. [2] [3] [1]

If you want, I can turn this into a search-ready annotated bibliography next: a tighter list of likely relevant papers/posts to inspect manually, sorted by your five categories.



1. <https://arxiv.org/html/2605.11746v1>
2. <https://arxiv.org/html/2602.00449v1>
3. <https://arxiv.org/html/2509.23676v1>
4. <https://dl.acm.org/doi/full/10.1145/3787104>
5. <https://arxiv.org/html/2502.10835v1>

6. <https://www.alignmentforum.org/posts/gtLLBhzQTG6nKTeCZ/attribution-patching-activation-patching-at-industrial-scale>
7. <https://arxiv.org/html/2603.22816v3>
8. <https://arxiv.org/html/2605.07284v1>
9. <https://arxiv.org/html/2604.00778v1>
10. <https://www.alignmentforum.org/posts/r2yTwkGt3kbQG2mXi/axrp-episode-19-mechanistic-interpretability-with-neel-nanda>
11. <https://arxiv.org/html/2604.12426v1>
12. <https://arxiv.org/pdf/2605.12991.pdf>
13. <https://www.anthropic.com/claude-sonnet-4-5-system-card>
14. <https://www-cdn.anthropic.com/Odd865075ad3132672ee0ab40b05a53f14cf5288.pdf>
15. <https://www.alignmentforum.org/posts/jP9KDyMkchuv6tHwm/how-to-become-a-mechanistic-interpretability-researcher>
16. <https://www.alignmentforum.org/posts/WzHPpMz2kRongsA7q/what-is-the-functional-role-of-sae-errors>
17. <https://www.alignmentforum.org/posts/dqSwccGTWyBgxrR58/turntrout-s-shortform-feed>
18. Screenshot2026-05-14-at-8.49.22-PM.jpg
19. <https://transformer-circuits.pub/2026/nla/>
20. <https://arxiv.org/pdf/2605.11746.pdf>
21. <https://ucsc-vlaa.github.io/publications.html>
22. <https://papers.lunadong.com/area/reasoning>
23. <https://www.alignmentforum.org/posts/YGAimivLxycZcqRFR/can-we-interpret-latent-reasoning-using-current-mechanistic>